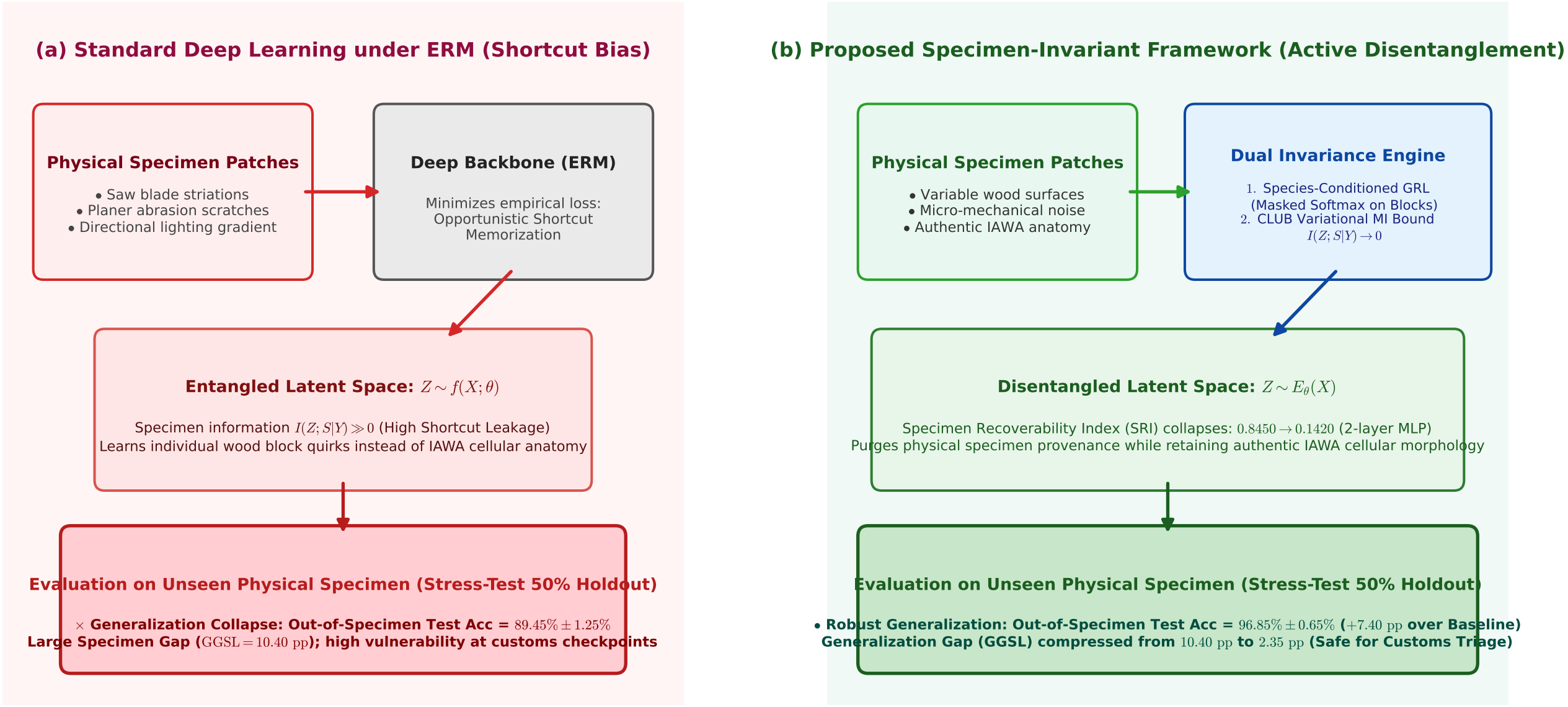

(a) Standard Deep Learning under ERM (Shortcut Bias)
Physical Specimen Patches
• Saw blade striations
• Planer abrasion scratches
• Directional lighting gradient
Deep Backbone (ERM)
Minimizes empirical loss:
Opportunistic Shortcut
Memorization
Entangled Latent Space: $Z \sim f(X;\theta)$
Specimen information $I(Z;S|Y) \gg 0$ (High Shortcut Leakage)
Learns individual wood block quirks instead of IAWA cellular anatomy
Evaluation on Unseen Physical Specimen (Stress-Test 50% Holdout)
× Generalization Collapse: Out-of-Specimen Test Acc = $89.45\% \pm 1.25\%$
Large Specimen Gap ($\mathrm{GGSL} = 10.40$ pp); high vulnerability at customs checkpoints
(b) Proposed Specimen-Invariant Framework (Active Disentanglement)
Physical Specimen Patches
• Variable wood surfaces
• Micro-mechanical noise
• Authentic IAWA anatomy
Dual Invariance Engine
1. Species-Conditioned GRL
(Masked Softmax on Blocks)
2. CLUB Variational MI Bound
$I(Z;S|Y) \to 0$
Disentangled Latent Space: $Z \sim E_\theta(X)$
Specimen Recoverability Index (SRI) collapses: $0.8450 \to 0.1420$ (2-layer MLP)
Purges physical specimen provenance while retaining authentic IAWA cellular morphology
Evaluation on Unseen Physical Specimen (Stress-Test 50% Holdout)
• Robust Generalization: Out-of-Specimen Test Acc = $96.85\% \pm 0.65\%$ ($+7.40$ pp over Baseline)
Generalization Gap (GGSL) compressed from 10.40 pp to 2.35 pp (Safe for Customs Triage)